Skills Review

January 22, 2020

Objectives

1. For a bond, compute the
   - price and
   - (implied) yield
2. For a stock, compute the
   - price,
   - (implied) expected dividend growth rate

Treasuries

A T-year Treasury makes semi-annual payments.

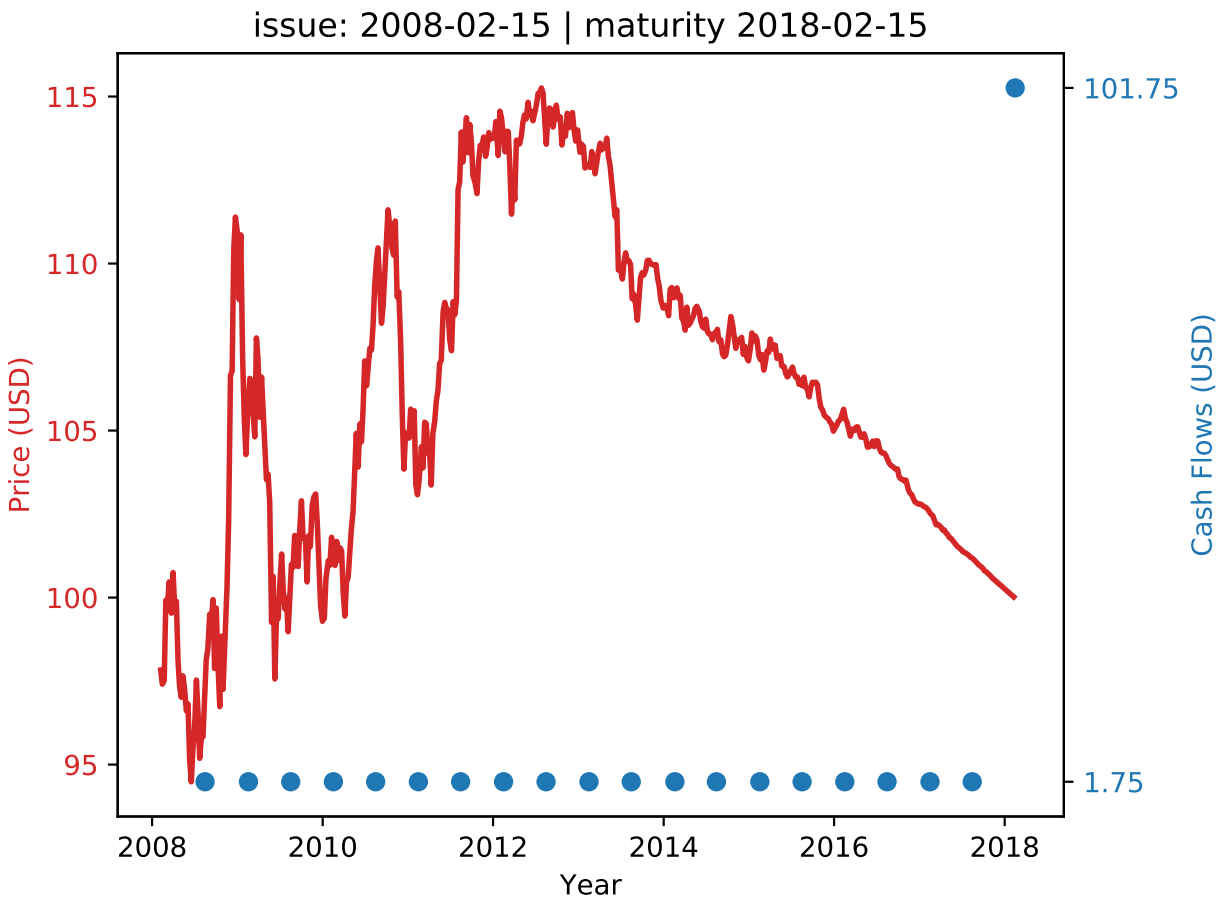
0	1	2	$\cdots$	$2T$
$-P$	$cF/2$	$cF/2$	$\cdots$	$cF/2 + F$

where $c \geq 0$ is the coupon rate (APR)

$$P \overset{\text{should}}{=} \frac{cF/2}{(1+y/2)} + \frac{cF/2}{(1+y/2)^2} + \frac{cF/2}{(1+y/2)^3} + \cdots + \frac{cF/2}{(1+y/2)^{2T}} + \frac{F}{(1+y/2)^{2T}}$$
$$= \frac{cF/2}{y/2}\left(1 - \frac{1}{(1+y/2)^{2T}}\right) + \frac{F}{(1+y/2)^{2T}}$$

where y is the yield (APR).

From the Data



Treasury Checklist

- ✓ Yields and coupon rates are quoted as APRs, so divide them by the number of payments per year to get the per-period rate
- ✓ Multiply the number of years by the number of payments per year to get the number of periods
- ✓ The future value is the face value
- ✓ Remember: if the coupon rate is greater than (less than) the yield, the price is greater than (less than) the face value
- ✓ **Use four decimal places!**

Example

On 2010-06-01, the yield on a 100 USD 5.0-year Treasury that matured on 2012-05-31 was 0.7124%. It paid a 4.75% coupon. What should it have traded for? (Hint: the number of payments up to and including maturity is 4.0)

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$$
\begin{aligned}
P_0 \overset{\text{should}}{=} & -\text{PV}(\\
& \quad \text{N} = 4.0, \\
& \quad \text{I/Y} = 0.7124\%/2, \\
& \quad \text{PMT} = (4.75\% \times 100)/2, \\
& \quad \text{FV} = 100) \\
= & \ 108.0038 \text{ USD}
\end{aligned}
$$

Example

On 2010-04-30, the yield on a 100 USD 5.0-year Treasury that matured on 2012-04-30 was 0.9183%. It paid a 4.5% coupon. What should it have traded for? (Hint: the number of payments up to and including maturity is 4.0)

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& \quad \text{I/Y} = 0.9183\%/2, \\
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& \quad \text{FV} = 100) \\
= & \ 107.0819 \text{ USD}
\end{aligned}
$$

Example

On 2016-02-29, a 7.0-year Treasury that matured on 2017-08-31 traded for 101.668 USD. It paid a 1.875% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 3.0)

Example

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\begin{aligned}
y_0 &\overset{\text{should}}{=} \text{RATE}(\\
&\qquad \text{N} = 3.0,\\
&\qquad \text{PV} = -101.668,\\
&\qquad \text{PMT} = (1.875\% \times 100)/2,\\
&\qquad \text{FV} = 100)\\
&= 0.7546\%
\end{aligned}
$$

Example

On 2010-07-15, a 3.0-year Treasury that matured on 2012-07-15 traded for 101.7969 USD. It paid a 1.5% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 4.0)

Example

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y_0 &\overset{\text{should}}{=} \text{RATE}(\\
&\qquad \text{N} = 4.0,\\
&\qquad \text{PV} = -101.7969,\\
&\qquad \text{PMT} = (1.5\% \times 100)/2,\\
&\qquad \text{FV} = 100)\\
&= 0.5949\%
\end{aligned}
$$

Stocks

- P_1 is the expected price in one period (usually one quarter)
- r is the required total rate of return (usually quarterly)
- D_1 is the expected dividend
- The price <u>should</u> be

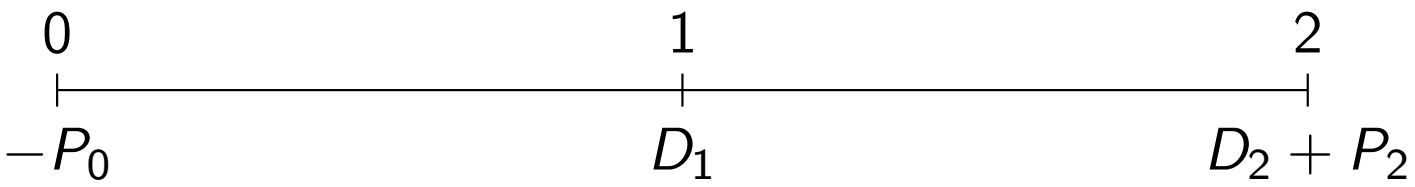
$$P_0 \overset{\text{should}}{=} \frac{D_1 + P_1}{1 + r}.$$

Didend Yields, Capital Gains, and Total Returns

$$r \overset{\text{should}}{=} \frac{D_1 + P_1}{P_0} - 1 = \underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- The dividend yield is the expected annual dividend of the stock divided by its current price
- The capital gain rate is the percent by which the stock price increases from the current period to the next
- The total return is the sum of the dividend yield and the capital gain rate

Stocks



- The prices should be

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} \text{ and } P_1 \overset{\text{should}}{=} \frac{D_2 + P_2}{1+r}.$$

- Does the two-year investor value the stock differently than the one-ear investor? **No!**

$$P_0 = \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} = \frac{D_1}{1+r} + \frac{P_1}{1+r} = \frac{D_1 + P_1}{1+r}$$

The Dividend Discount Model

- It doesn't matter whether the investor wants to hold the stock for one quarter, two quarters, or infinitely many quarters: she is always willing to pay the same price P_0
- In particular, the investor who holds the stock for infinitely many periods is willing to pay

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \cdots$$

  for the stock
- Implication: we don't need to know the prices P_1, P_2, ..., just the dividends D_1, D_2, D_3, ...

Constant Dividend Growth

Simple model: dividends grow at a constant rate g

0	1	2	3	$\cdots$
$-P_0$	$(1+g)D_0$	$(1+g)^2 D_0$	$(1+g)^3 D_0$	$\cdots$

where D_0 is the dividend that was <u>just</u> paid

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{(1+r)} + \frac{(1+g)^2 D_0}{(1+r)^2} + \frac{(1+g)^3 D_0}{(1+r)^3} + \cdots = \frac{(1+g)D_0}{r-g}$$

Constant Dividend Growth

- The Dividend Discount Model says that

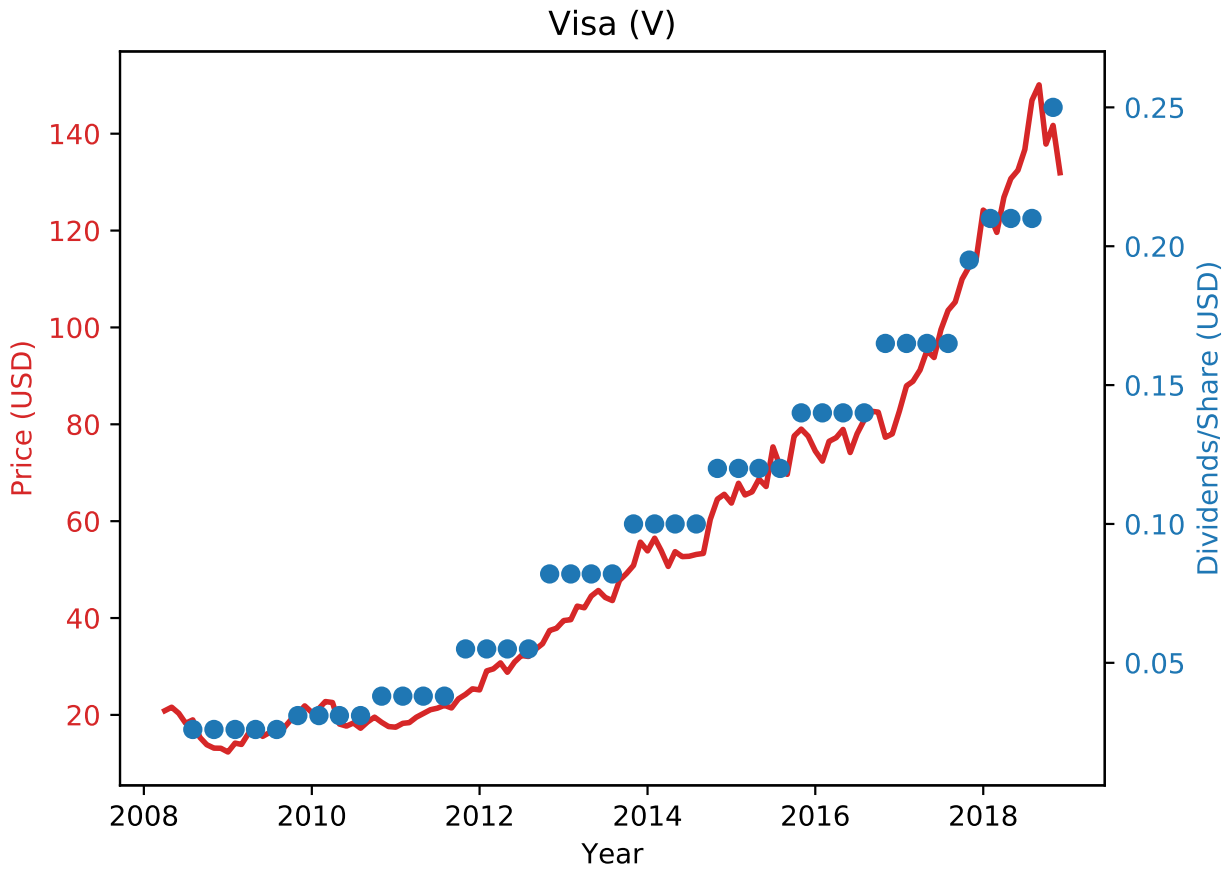
$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{D_1}{r-g}$$

- Awesome interpretation!

$$\overbrace{\frac{D_1}{P_0} + g}^{\text{Specific Formula}} \overset{\text{should}}{=} r \overset{\text{should}}{=} \overbrace{\underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}}^{\text{General Formula}}$$

- In this case, the <u>growth rate</u> g is the <u>capital gain rate</u>

From the Data



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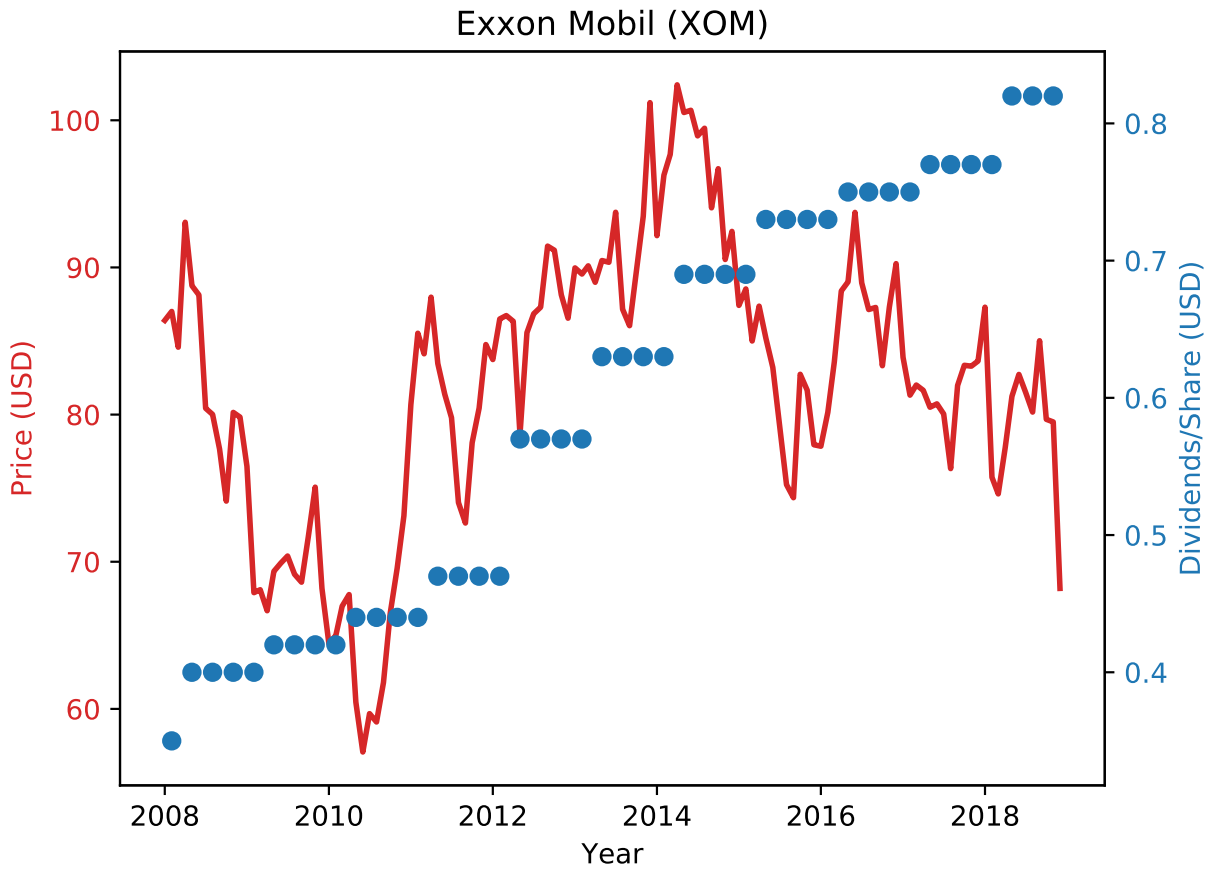

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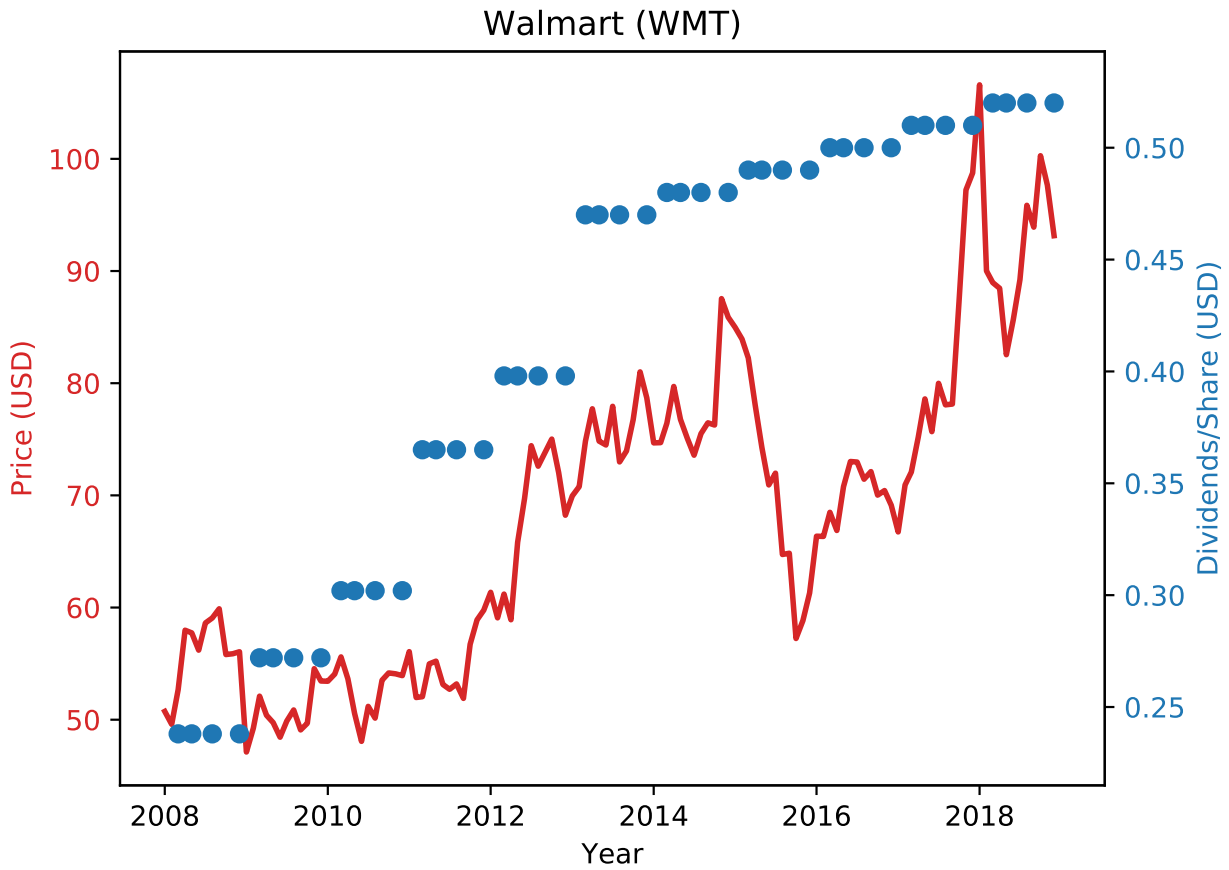

From the Data

Does the constant dividend growth assumption comport with reality?

- Verizon's dividends are (generally) constant across the four quarters in each year, but increase from the fourth quarter of one year to the first quarter of the next
- Exxon's dividends follow a similar pattern to those of Verizon, despite the much greater volatility in its stock price
- Walmart's dividends follow a similar pattern to those of Verizon and Exxon, but the annual growth in the dividend rate seems to kink in 2013

There are some issues...

Stock Checklist

- ✓ The required rate of total return should have the same horizon as that of the dividends, so if returns are quoted in months, say r_m, compute the quarterly return as follows:

$$r_q = (1 + r_m)^3 - 1 \tag{1}$$

- ✓ The growth rate should also have the same horizon as that of the dividends
- ✓ **Use four decimal places!**

Example: Stock Price

Chevron (CVX) requires a quarterly total return of 1.671%. On 2010-05-01, it paid a dividend of 0.72 USD per share (Chevron's dividends had been growning at an average quarterly rate of 1.439%). Based on this information, what should Chevron's stock have traded for? (For reference, it actually traded for 73.87 USD.)

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$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{0.72 + 1.439\% \times 0.72}{1.671\% - 1.439\%} = 314.811 \text{ USD}$$

Example: Stock Price

JPMorgan Chase (JPM) requires a quarterly total return of 2.532%. On 2017-04-01, it paid a dividend of 0.5 USD per share (JPMorgan Chase's dividends had been growning at an average quarterly rate of 1.026%). Based on this information, what should JPMorgan Chase's stock have traded for? (For reference, it actually traded for 87.0 USD.)

Example: Stock Price

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$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{0.5 + 1.026\% \times 0.5}{2.532\% - 1.026\%} = 33.541 \text{ USD}$$

Example: Implied Growth Rate

Goldman Sachs (GS) requires a quarterly total return of 2.712%. On 2015-05-01, it paid a dividend of 0.65 USD per share, after which it traded for 206.19 USD. What was the implied quarterly dividend growth rate? (For reference, our estimate would be 2.021%).

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$$g \overset{\text{should}}{=} \frac{rP_0 - D_0}{P_0 + D_0} = \frac{2.712\% \times 206.19 - 0.65}{206.19 + 0.65} = 2.389\%$$

Example: Implied Growth Rate

United Technologies (UTX) requires a quarterly total return of 2.136%. On 2015-02-01, it paid a dividend of 0.64 USD per share, after which it traded for 121.91 USD. What was the implied quarterly dividend growth rate? (For reference, our estimate would be 2.054%).

Example: Implied Growth Rate

United Technologies (UTX) requires a quarterly total return of 2.136%. On 2015-02-01, it paid a dividend of 0.64 USD per share, after which it traded for 121.91 USD. What was the implied quarterly dividend growth rate? (For reference, our estimate would be 2.054%).

$$g \overset{\text{should}}{=} \frac{rP_0 - D_0}{P_0 + D_0} = \frac{2.136\% \times 121.91 - 0.64}{121.91 + 0.64} = 1.603\%$$

Example: Dividend Yield

Merck (MRK) requires a quarterly total return of 1.265%. On 2015-12-01, it had an implied quarterly dividend growth rate of 0.391%. Under the assumption of constant dividend growth, what was its implied dividend yield?

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$$\text{dividend yield} = r - g = 1.265\% - 0.391\% = 0.874\%$$

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$$\text{dividend yield} = r - g = 1.265\% - 0.469\% = 0.796\%$$